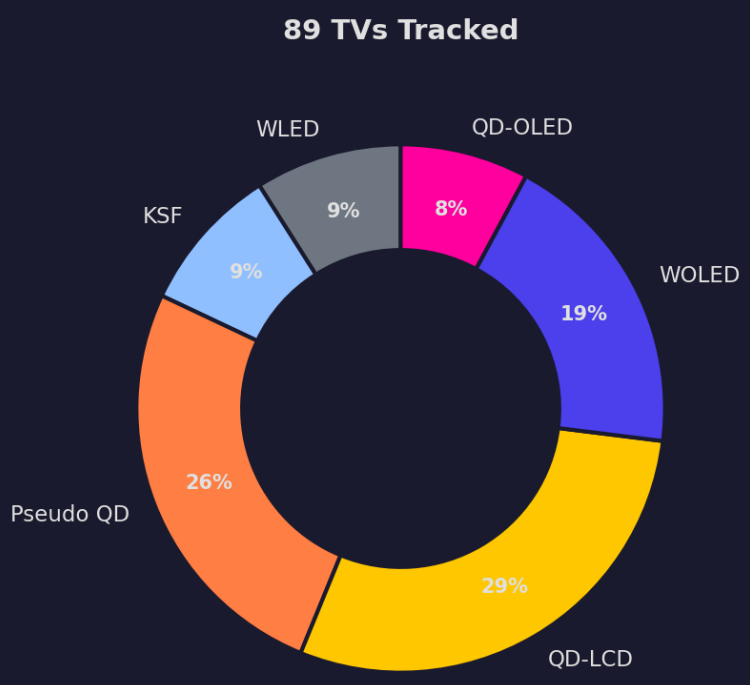
Display Technology Intelligence Report

May 2026

89 4K TVs tracked | 74 with pricing | 6 technologies | 3 8K excluded



1. Display Technology Overview

Section 1: Display Technology Overview -- May 2026

1.1 Catalog Changes

The May 2026 database holds 89 TVs across six display technologies, with two new entries and zero removals this month. The Hisense U7SG (QD-LCD) joins at a mixed-usage score of 8.3, and the Samsung S90H OLED (WOLED) debuts at 8.5 -- both landing well above their respective technology category averages of 7.9 and 8.4. The absence of any removals signals a stable, additive cycle: manufacturers are releasing new SKUs without retiring existing lines, which is consistent with the mid-year refresh pattern seen across premium and upper-mid tiers.

The two additions reinforce divergent strategic bets. Hisense continues to push QD-LCD further up the performance ladder, while Samsung's new WOLED entry competes directly against its own QD-OLED lineup rather than ceding the organic panel segment entirely to LG. With the catalog now spanning 89 models, the data set is large enough that single-model additions move category averages by less than a tenth of a point -- a sign of statistical maturity that makes outliers easier to identify and genuine trend shifts harder to fake.

The catalog's zero-removal month is likely temporary; expect Q3 to see legacy WLED and entry-tier KSF models age out as shelf space consolidates around higher-margin QD and OLED tiers.

1.2 Performance Snapshot by Technology

QD-OLED leads every scoring category in May 2026, posting a mixed-usage mean of 8.6, a home theater mean of 8.9, and a gaming mean of 8.9 -- with a floor of 8.2 on mixed usage, meaning even the weakest QD-OLED in the database outperforms the average QD-LCD (7.9) by a meaningful margin. WOLED sits just behind in mixed usage at 8.4 but edges closer to QD-OLED in gaming (8.7 vs. 8.9) and home theater (8.6 vs. 8.9), reflecting LG's continued improvements to panel brightness and tone mapping. The gap between these two OLED architectures is narrowing, but QD-OLED's spectral advantage keeps it ahead.

The LCD tiers tell a sharper story of stratification. QD-LCD's mixed-usage mean of 7.9 sits 1.4 points above Pseudo QD and KSF, both of which tie at 6.5. That tie is misleading -- Pseudo QD's ceiling reaches 8.3 in mixed usage (matched by the best performers in the segment, including potentially the new Hisense U7SG at exactly that mark), whereas KSF tops out at 7.6. WLED occupies the basement at a mixed-usage mean of 5.2, with a maximum of just 5.6, confirming it as a technology in terminal decline within the premium evaluation set.

The 1.3-point spread between QD-OLED's mean and WLED's mean on a 10-point scale represents a

genuine experiential chasm, not a marginal tuning difference -- and it will define retailer tiering conversations through the holiday cycle.

1.3 FWHM Spectral Analysis

Full-width at half-maximum (FWHM) figures measure how narrowly a display's backlight or emitter produces each primary color -- tighter values mean purer, more saturated output with less spectral bleed into adjacent primaries. QD-OLED leads on green purity with a mean FWHM of 27.8 nm, followed closely by QD-LCD at 29.7 nm. These two technologies effectively share the top tier for green spectral precision, separating themselves from every other architecture by a wide margin. On red, QD-OLED widens slightly to 37.1 nm while QD-LCD holds tight at 29.6 nm -- the one dimension where QD-LCD's filtered approach actually rivals OLED-based color.

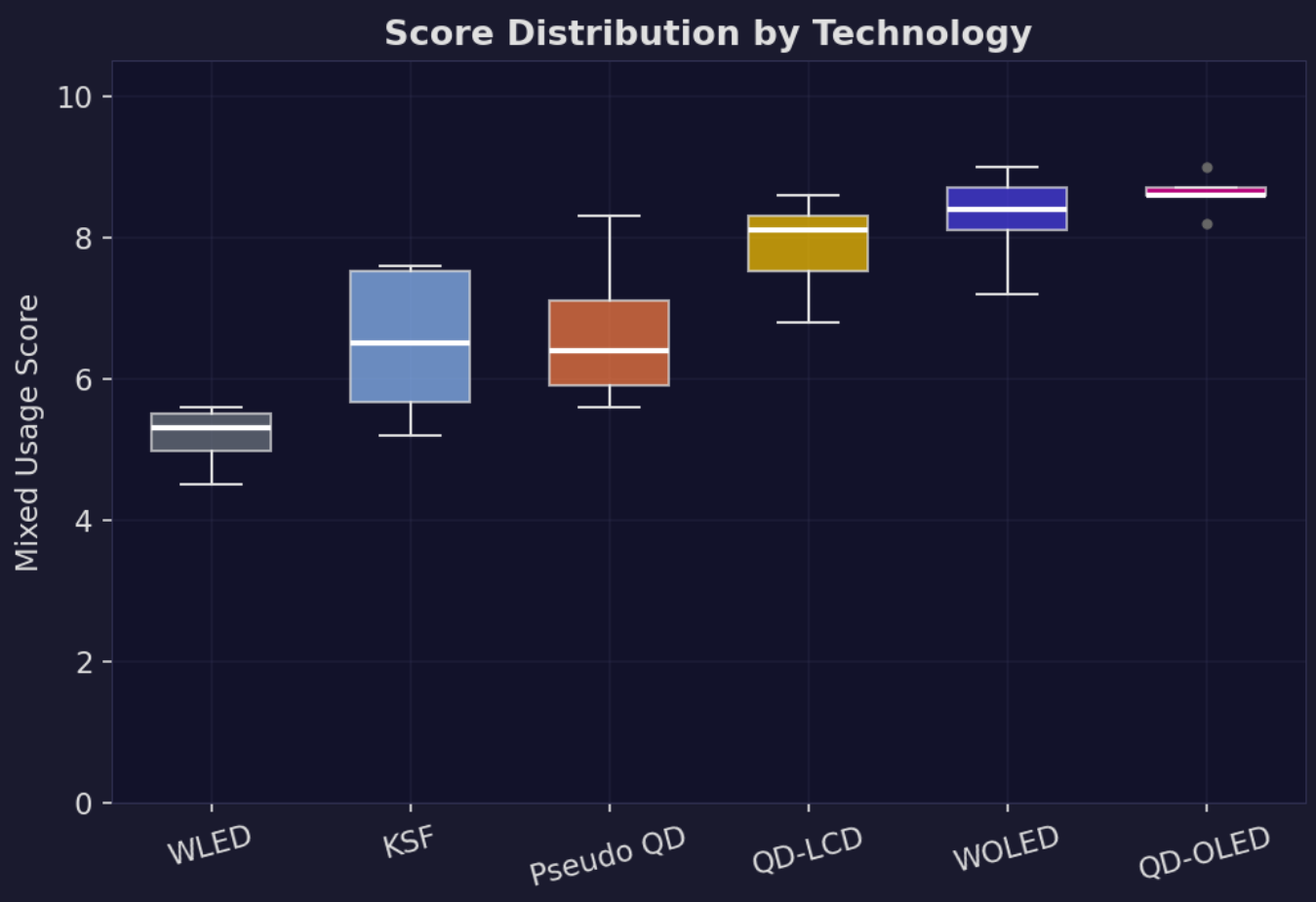
The legacy technologies show dramatically broader spectra. WLED measures 70.9 nm on green and 78.7 nm on red -- values more than double those of QD-OLED -- which directly explains its scoring deficit. WOLED surprises on the downside, posting the widest green FWHM in the entire dataset at 83.7 nm, a consequence of its white-emitter-plus-color-filter architecture. WOLED compensates through brightness, black level, and per-pixel dimming rather than spectral purity, which is why its scores remain competitive despite this optical disadvantage. KSF delivers an unusually narrow red at 9.1 nm -- the tightest red peak in the dataset by a significant margin -- though its green at 42.8 nm is less impressive, reflecting the phosphor technology's asymmetric strengths.

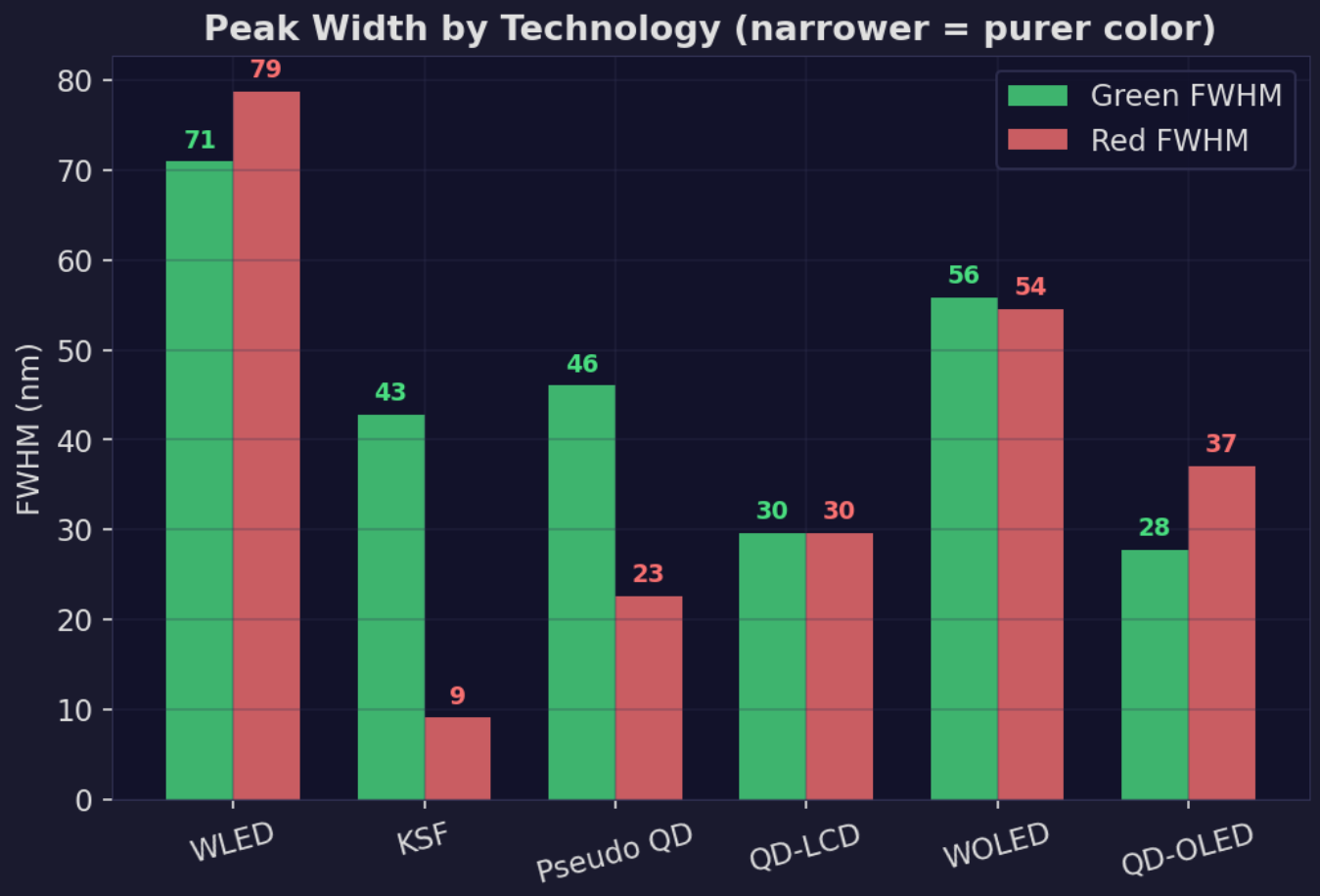
As QD material quality continues to improve, the already-thin spectral gap between QD-OLED and QD-LCD will become the central battleground for color volume claims in marketing and spec sheets.

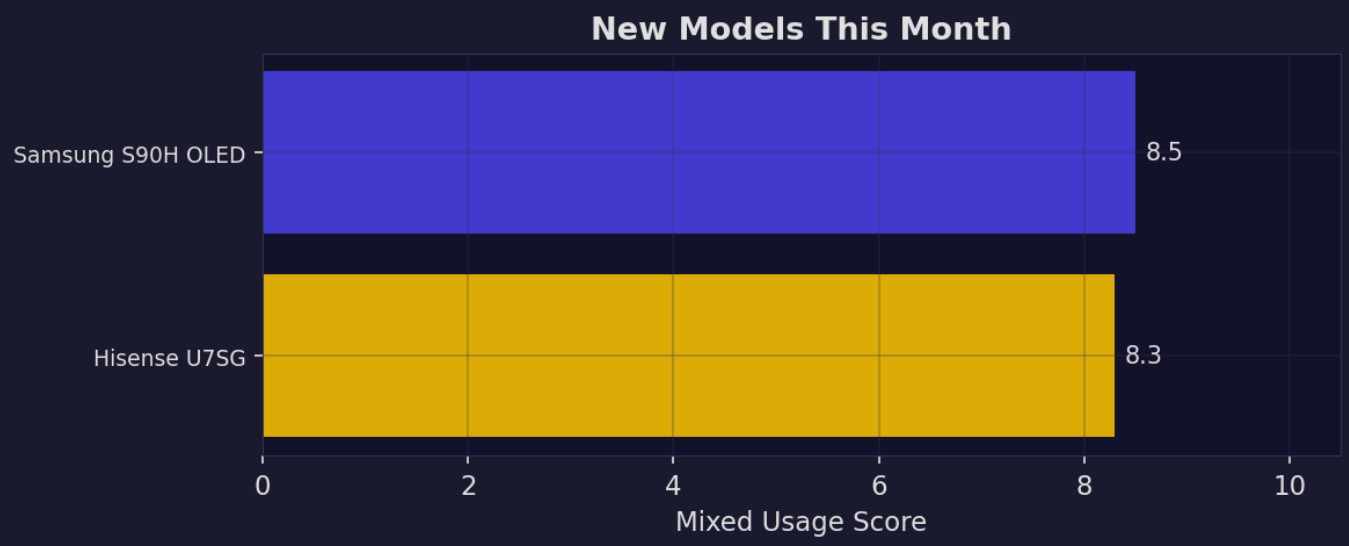
1.4 Model Year Cohort Comparison

The 2026 cohort is small -- 6 models against 41 from 2025 -- so averages should be read directionally rather than as settled benchmarks. That said, the early 2026 numbers are striking: a mixed-usage mean of 8.6 and a gaming mean of 8.8 against 2025's 7.3 across both categories. That is a 1.3-point year-over-year jump on mixed usage and a 1.5-point jump on gaming, suggesting the 2026 entrants are concentrated at the premium end of the market rather than distributed across tiers. The Samsung S90H OLED at 8.5 and the positioning of other early-cycle additions confirm that manufacturers are leading the year with flagship hardware.

The caveat is selection bias: six models cannot represent a full year-class, and the 2025 cohort of 41 spans entry-level through ultra-premium, pulling its mean toward the middle. As mid-range 2


Score Distribution by Technology
Mixed Usage Score
0
2
4
6
8
10
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED


Peak Width by Technology (narrower = purer color)
FWHM (nm)
0
10
20
30
40
50
60
70
80
Green FWHM
Red FWHM
71
79
43
9
46
23
30
30
56
54
28
37
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED


New Models This Month
Samsung S90H OLED
8.5
Hisense U7SG
8.3
0
2
4
6
8
10
Mixed Usage Score

2. Pricing Intelligence

Section 2: Pricing Intelligence

2.1 Current Pricing Landscape

Across 74 priced models tracked in May 2026, the TV market stratifies cleanly into three pricing tiers that map almost perfectly onto display technology. WLED sits at the base with a median price of $458, while KSF and Pseudo QD occupy a middle band at $1,252 and $724 respectively. The premium segment begins sharply with QD-LCD at $1,809 and escalates through WOLED at $1,889 to QD-OLED at the top of the stack at $2,130.

The per-square-meter lens sharpens those gaps considerably. WLED delivers screen area at $576/m2, establishing the commodity floor. QD-OLED commands $2,495/m2 -- a figure that reflects not just panel cost but the concentrated positioning of large-screen flagship SKUs in that category. WOLED sits just below at $2,357/m2, and QD-LCD lands at $2,073/m2, making all three premium emissive and quantum dot technologies cluster within a roughly $400/m2 band at the top while the LCD tiers spread across the remaining $1,500 of headroom beneath them.

The spread between the cheapest and most expensive technology on a per-area basis is 333% -- QD-OLED carries a $1,919/m2 premium over WLED. That number defines the ceiling of what the market currently tolerates for best-in-class picture quality, and it has not compressed meaningfully in the trailing 12-week window.

2.2 Price Trends

Twelve weeks of pricing data provides a statistically meaningful trend window, and the headline is that most technologies are holding or drifting modestly -- with one significant outlier. WLED ticked up 2.3% to $576/m2 from $563/m2, consistent with tariff-related input cost pressure rather than any demand-driven repricing. QD-LCD and QD-OLED moved +1.0% and +4.3% respectively, suggesting stable but gradually firming premium positioning as 2026 model year inventory replaces closeout 2025 units.

The anomaly is WOLED, which climbed 18.2% over the period -- from $1,994/m2 to $2,357/m2. That magnitude of movement in 12 weeks is not promotional noise; it reflects a structural repricing, most likely driven by LG Display's reduced panel allocation to third-party brands as the company prioritizes its own retail channel and defends margin on a technology that faces intensifying QD-OLED competition. KSF moved in the opposite direction, declining 1.7% to $1,160/m2 -- a modest but directionally meaningful concession as TCL pushes volume on phosphor-enhanced LCD.

Pseudo QD posted a +6.0% move to $878/m2, which on the surface looks like strength but more likely

reflects a mix shift toward larger-panel SKUs as entry Pseudo QD models exit the tracked set. The absolute dollar medians should be read alongside the per-area figures to separate genuine price movement from size-mix effects.

The WOLED repricing deserves close monitoring -- if it holds through June, it will begin to erode that technology's value proposition relative to QD-OLED and accelerate share rotation toward Samsung and Sony's quantum dot panels.

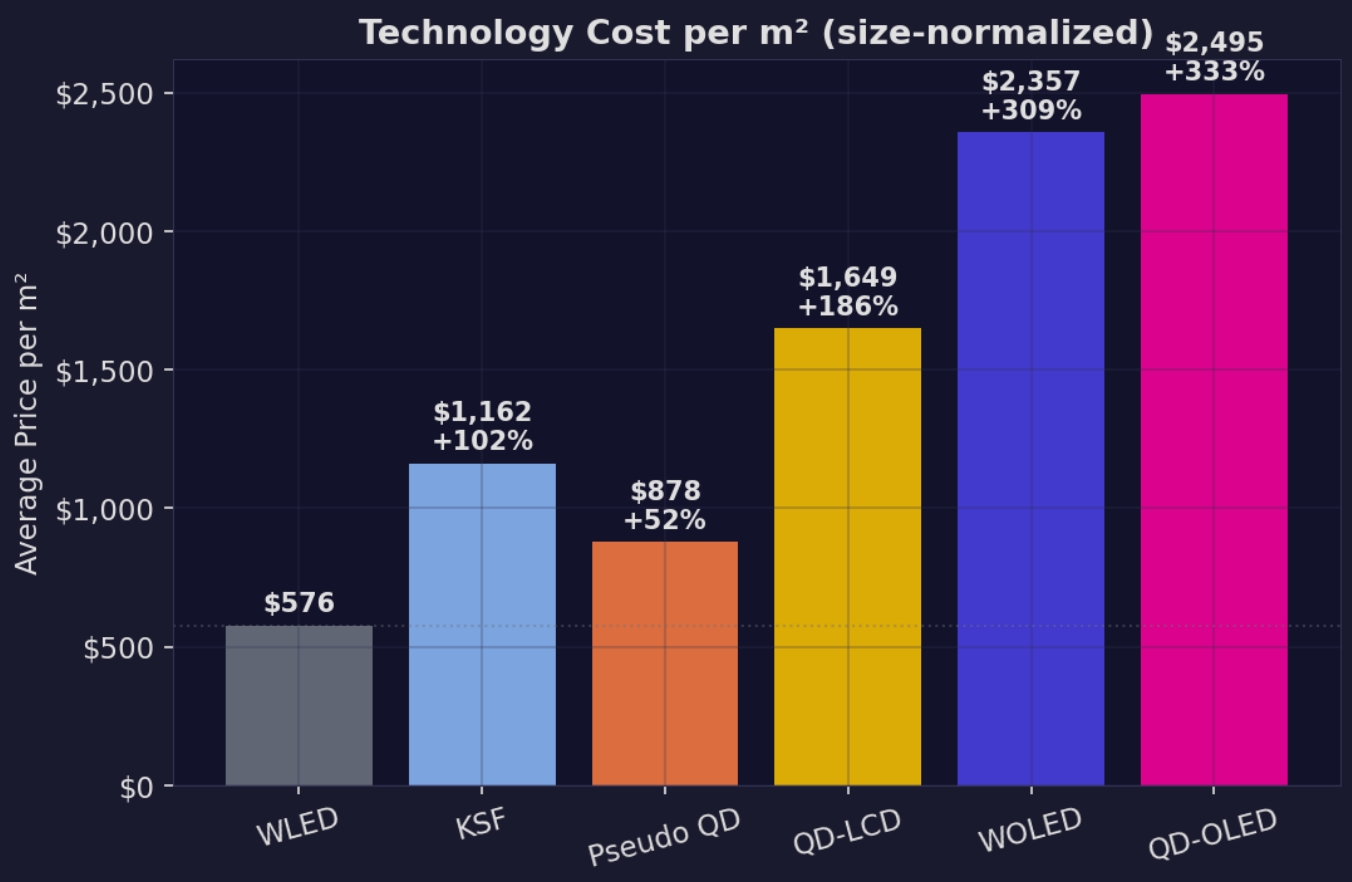
2.3 Value Analysis

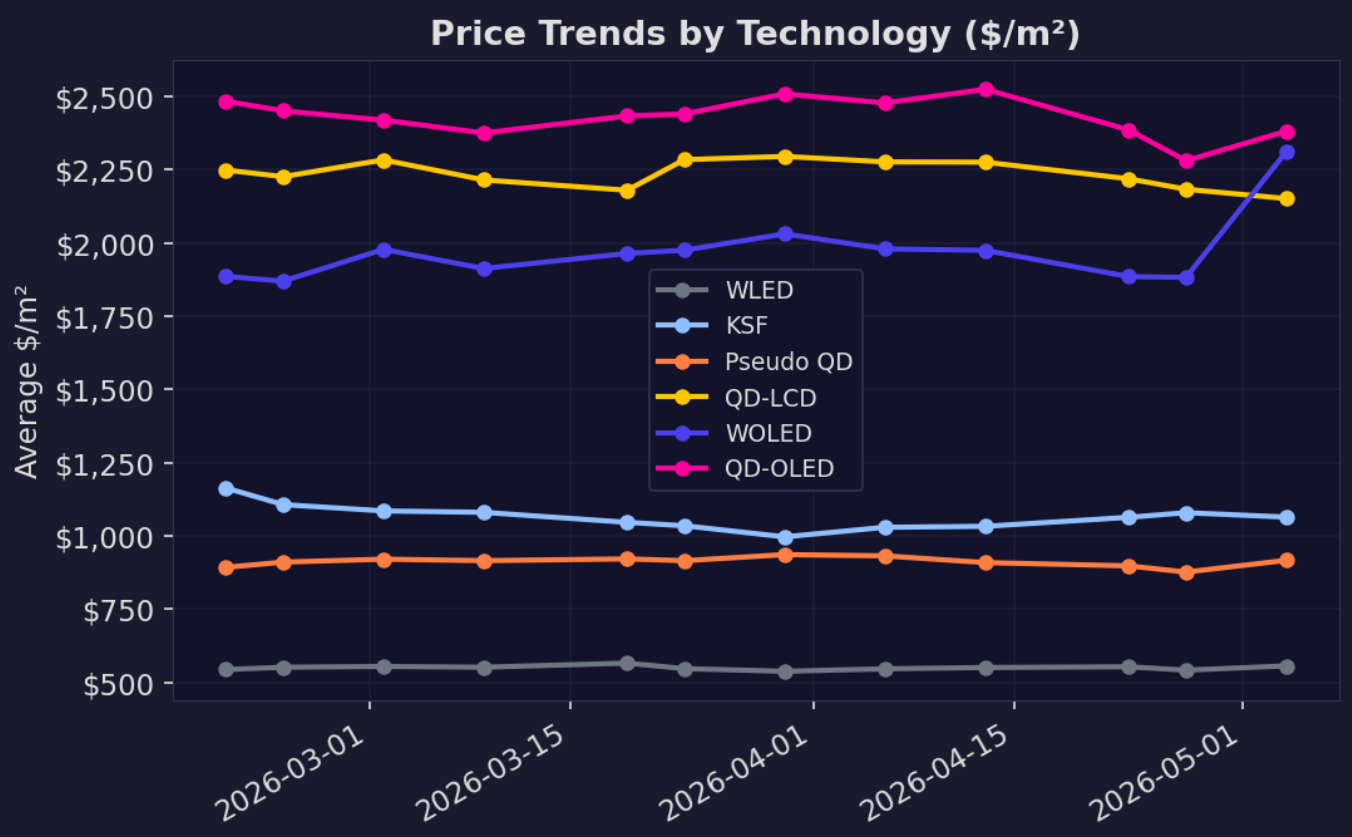
Performance-per-dollar is where the mid-tier LCD technologies make their clearest argument. The TCL S551G, a KSF-based panel, leads the tracked set with a mixed-usage score of 5.9 at just $370, translating to $63 per performance point -- the sharpest value ratio in the dataset. The TCL Q651G (Pseudo QD, $380, score 5.8) follows at $66/point, and the Hisense U6N (Pseudo QD, $451, score 6.7) rounds out the top three at $67/point. Two WLED models -- the Panasonic W70B at $69/point and the Samsung U8000F at $72/point -- complete the top five, confirming that the value tier is entirely an LCD story in May 2026.

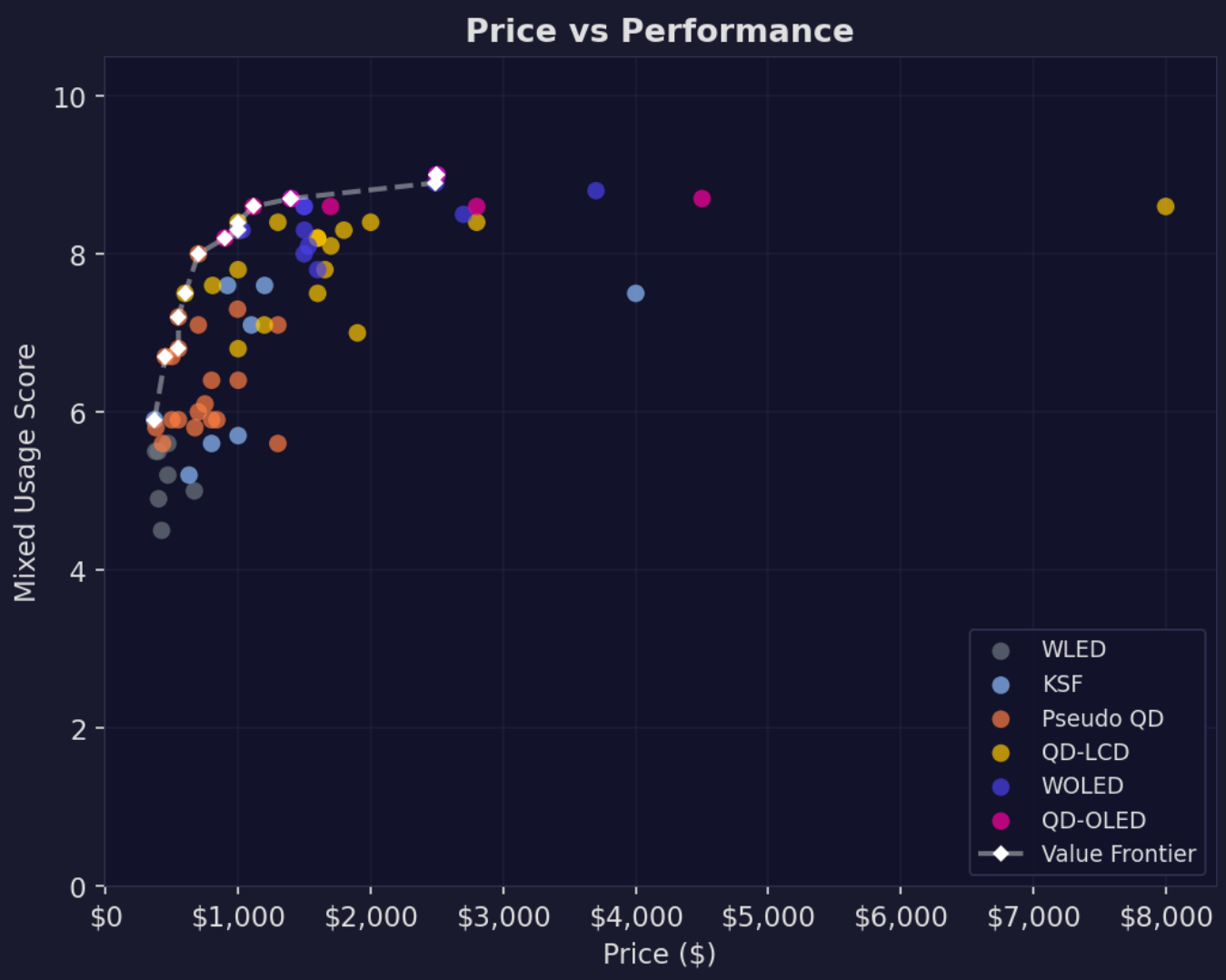
None of the premium emissive or QD-LCD technologies crack the top five on dollar-per-point efficiency. QD-OLED delivers superior absolute scores, but its $2,495/m2 pricing means buyers are paying for a step-change in HDR performance and viewing-angle fidelity that the value metrics don't fully capture in a blended usage score. The gap between the $63/point efficiency of the S551G and the implicit cost-per-point of a flagship QD-OLED is likely 3 - 4x -- a premium that targets a fundamentally different buyer.

The practical implication for retail positioning is that KSF is punching above its weight on value. TCL has priced the S551G below the KSF category median of $1,252 -- suggesting the broader KSF installed base skews toward larger-format configurations, and that there is room to expand KSF's value narrative at 55" and below without cannibalizing the mid-range Pseudo QD lineup.

Brands that close the per-point efficiency gap between Pseudo QD and WLED -- currently just $3 - 4/point -- will own the most contested battleground in the market heading into the second half of 2026.


Technology Cost per m² (size-normalized)
Average Price per m²
$2,500
$2,000
$1,500
$1,000
$500
$0
$576
$1,162
+102%
$878
+52%
$1,649
+186%
$2,357
+309%
$2,495
+333%
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED


Price Trends by Technology ($/m²)
Average $/m²
$2,500
$2,250
$2,000
$1,750
$1,500
$1,250
$1,000
$750
$500
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED
2026-03-01
2026-03-15
2026-04-01
2026-04-15
2026-05-01


Price vs Performance
Mixed Usage Score
Price ($)
0
2
4
6
8
10
$0
$1,000
$2,000
$3,000
$4,000
$5,000
$6,000
$7,000
$8,000
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED
Value Frontier

3. Macro & Industry Context

Section 3: Macro & Industry Context

The Display Technology Landscape, May 2026

Our 89-television database tells a story of a market that has fragmented far beyond the simple LCD-vs-OLED binary that defined the early 2020s. Six distinct display technologies now compete for shelf space, and the distribution is telling: QD-LCD leads with 26 panels tracked, followed by Pseudo QD at 23 -- meaning nearly half the database is quantum-dot-adjacent LCD, a technology family that spans an enormous performance and price range. True self-emissive technologies -- WOLED (17), QD-OLED (7) -- account for just 27% of tracked models, yet command a disproportionate share of review attention and margin. The bottom tier, WLED (8) and KSF (8), persists largely because the sub-$400 mass market remains enormous, and basic phosphor LCD is still the cheapest way to fill a 65-inch frame. Samsung's 25-model presence dominates the database, with Hisense and TCL together representing 29 panels -- a clear signal that Chinese OEMs have moved decisively from budget-only positioning into the full performance stack.

Technology Differentiation: What the Tiers Actually Mean

The performance gap between technology classes is not a marketing abstraction -- it is measurable in the metrics that drive purchase decisions. QD-OLED, with only 7 models tracked, sits at the apex of color volume and motion clarity: quantum dots on a self-emissive substrate solve the brightness-versus-color-accuracy tradeoff that plagued first-generation OLEDs. WOLED's 17 models represent LG Display's incumbent advantage -- excellent black levels and wide viewing angles, but a white subpixel architecture that caps color saturation relative to QD-OLED. The practical ceiling for WOLED is higher peak brightness on 2026 tandem-stack panels, but the color volume disadvantage against QD-OLED at equivalent brightness persists. QD-LCD's 26 models span Sony's premium Mini-LED flagships through mid-tier Samsung and TCL entries -- the technology's core strength is peak HDR luminance, its core weakness is the local dimming halo that no backlight zone count has fully eliminated.

The Pseudo QD category at 23 models is the database's most commercially significant sleight of hand. These panels use KSF (potassium fluorosilicate) narrow red phosphor -- a legitimate improvement over basic WLED -- but are marketed under quantum dot branding by manufacturers seeking to occupy a premium price tier without quantum dot material costs. The delta between KSF and true QD-LCD in color gamut coverage is real and measurable on a spectrophotometer, even when it is invisible to a showroom shopper under store lighting.

Where the Industry Is Heading

Three technology vectors are converging on the 2027 - 2029 product cycle. Tandem OLED stacking -- two emissive layers in series -- has already moved from prototype to production, pushing WOLED brightness ceilings past 1,000 nits sustained and beginning to close the luminance gap with Mini-LED QD-LCD.

Perovskite quantum dots are the materials story that has circulated in research papers for four years and is now approaching manufacturable purity thresholds; if perovskite replaces cadmium-free QD in LCD backlights, the color volume of mid-tier QD-LCD panels increases substantially at lower BOM cost. MicroLED remains the technology the industry wants but cannot yet price for the living room -- panel-level yields and per-pixel driver costs keep 100-inch microLED above $10,000, making it a commercial installation and luxury residential product through at least 2028.

The actionable implication for a display technology team is this: the Pseudo QD segment's 23-model footprint represents the most vulnerable tier in the database -- it is the category most exposed to margin compression as true QD-LCD costs fall and perovskite materials reach volume pricing, and the brands that have built positioning there need a technology migration roadmap before that floor drops.

4. Year-over-Year Technology Trends

Section 4: Year-over-Year Technology Trends

4.1 Score Progression by Panel Technology

QD-LCD delivered the most dramatic quality improvement of any panel technology between the 2025 and 2026 model years, posting a +0.7-point gain in mixed-usage score -- rising from 7.7 to 8.4 across the reviewed sample. That is a substantial single-year jump for a mature LCD-adjacent platform, suggesting panel suppliers and TV manufacturers pushed meaningful engineering refinements into the quantum dot layer, local dimming algorithms, or both. The fact that QD-LCD's 2026 score of 8.4 now matches what WOLED scored in 2025 reframes the competitive landscape entirely.

WOLED also improved, but at a more measured pace -- its average mixed-usage score climbed +0.3 points, from 8.4 to 8.7. That incremental gain keeps WOLED ahead, but the cushion has compressed to just 0.3 points, down from a 0.7-point advantage one year ago. LG Display's WOLED manufacturing process is already highly optimized, so sustaining even modest annual score gains represents real engineering progress. The gap, however, is closing faster than the OLED camp likely anticipated.

The score convergence between these two technologies is the defining trend of the 2026 model year, and it puts the burden squarely on WOLED to differentiate on dimensions beyond raw mixed-usage performance before QD-LCD closes that remaining margin entirely.

4.2 Pricing Dynamics: Compression Below, Expansion Above

The pricing trajectories for these two technologies moved in opposite directions with unusual force. QD-LCD shed $341/m2 between model years, dropping from $1,515/m2 in 2025 to $1,174/m2 in 2026 -- a 22.5% price-per-area reduction in a single cycle. Part of this decline reflects natural panel commoditization, but the concurrent score *increase* means manufacturers are delivering more performance per dollar, not just cutting corners to hit lower price points. Retailers and brands leaning into the mid-premium segment will find QD-LCD a structurally more attractive platform in 2026 than it was 12 months ago.

WOLED moved the opposite direction with equal conviction. Average price-per-square-meter climbed from $2,483 to $3,248 -- a 30.8% increase, or roughly +$765/m2 year-over-year. That pricing expansion coincides with a contraction in the number of WOLED models in the reviewed sample, down from 6 in 2025 to 4 in 2026, which suggests the category is consolidating around larger, more premium SKUs rather than broadening its addressable market. Higher average screen area in the 2026 WOLED cohort would mechanically push $/m2 upward, but even accounting for size mix shifts, the directional signal is unmistakable: WOLED is moving upstream.

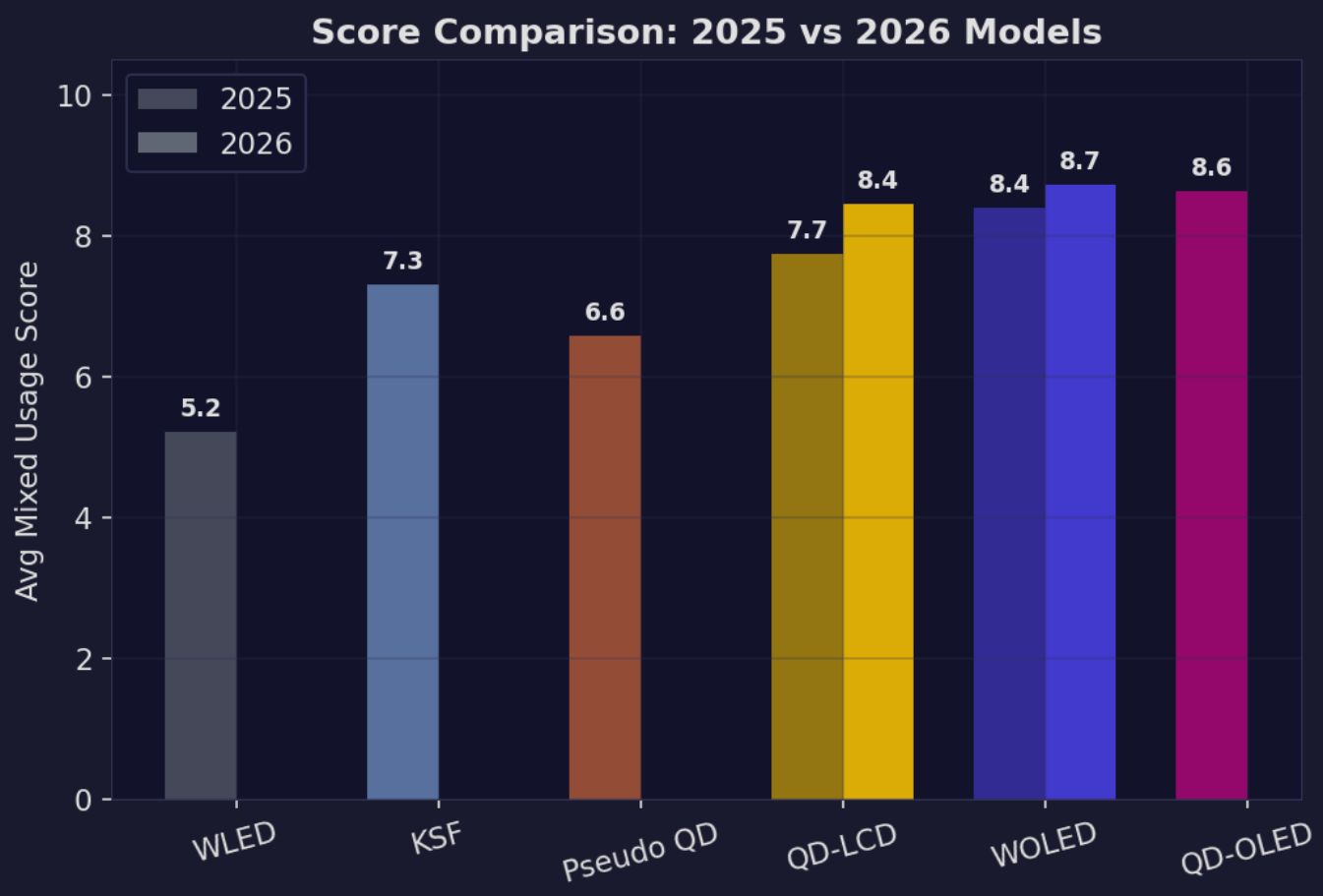
The pricing vectors are diverging at a rate that will stress the value narrative for WOLED at mid-tier screen sizes, and brands that straddle both panel technologies in their lineup need to recalibrate their price-ladder architecture now.

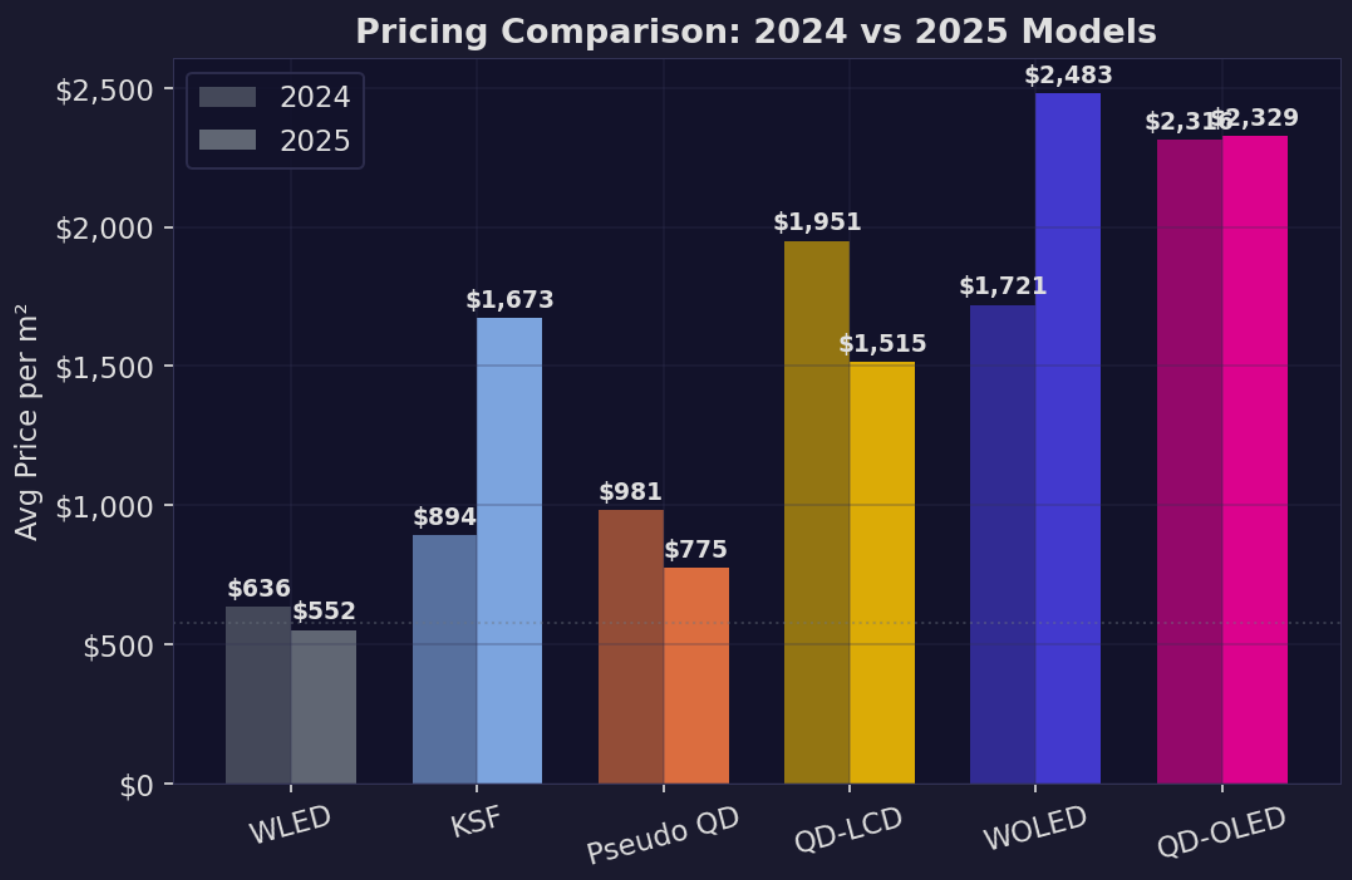
4.3 Competitive Positioning and Technology Trajectories

The combined effect of QD-LCD's score gains and price compression versus WOLED's score incrementalism and price expansion produces a measurably narrower competitive moat for OLED in 2026. A year ago, WOLED commanded a 0.7-point quality premium and a $968/m2 price premium. Today, that quality gap has shrunk to 0.3 points while the price gap has widened to $2,074/m2. Consumers paying the WOLED premium are now buying a smaller performance differential at a significantly higher cost.

The sample-size dynamics amplify this shift. QD-LCD's reviewed pool shrank from 12 to 2 models, meaning the 2026 QD-LCD scores reflect a more curated, higher-performing subset rather than a broad market average -- a sign that lower-performing QD-LCD SKUs are either being discontinued or have migrated to standard LCD categorization. WOLED's pool contraction from 6 to 4 models tells a different story: SKU rationalization toward large-screen flagships, not a widening of the technology's reach.

If QD-LCD's score trajectory holds through 2027 while WOLED pricing continues its upward drift, the OLED premium will need to be justified by attributes outside the mixed-usage score -- most likely in dark-room contrast, response time, or brand cachet -- or the technology risks ceding the upper-mid-market to a QD-LCD segment that is rapidly earning its place there.


Score Comparison: 2025 vs 2026 Models
2025
2026
Avg Mixed Usage Score
0
2
4
6
8
10
5.2
7.3
6.6
7.7
8.4
8.4
8.7
8.6
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED


Pricing Comparison: 2024 vs 2025 Models
2024
2025
Avg Price per m²
$2,500
$2,000
$1,500
$1,000
$500
$0
$636
$552
$894
$1,673
$981
$775
$1,951
$1,515
$1,721
$2,483
$2,316
$2,329
WLED
KSF
Pseudo QD
QD-LCD
WOLED
QD-OLED